Binocular Rivalry

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Abstract

In this Fall 2021 GaTech *PHYS 6268 Nonlinear Dynamics* course project, we explore the perceptual phenomenon of binocular rivalry, which occurs when two different images are presented simultaneously to each eye and perception alternates between the images. These perceptual alternations depend on various factors and are naturally conducive to a dynamical systems approach. Various phenomenological models have been constructed to explain the possible trajectories of brain activity that could underlie perceptual alternations. Those models revolve around established principles like neural competition and mutual inhibition. However, there are aspects of binocular rivalry that remain unexplained, such as where in the brain perceptual dominance is determined, and what role is played by the hierarchical organization of the visual system. In this project, we explore these questions by building a simple 2 dimensional model of binocular rivalry. We hope that, with this approach, we can explain what each parameter does and how it contributes to the behavior of the model.

1 Introduction

From the beginning, life on Earth has evolved alongside the rhythmic presence of the sun's warming light. As a consequence, biological organisms have adapted to manipulate this light for useful purposes. Plants perform photosynthesis to transform the energy of photons into nourishing food that sustains them. Single-celled organisms like *Halobacterium salinarum* have evolved to do a similar task, but through a different mechanism than photosynthesis. Upon exposure to light, the molecule retinal in this micro-organism's cell membrane goes through a process that ends in the production of chemical energy from light. The same molecule, retinal, initiates the process of vision in the photoreceptors of animals. Whereas the previous two examples in plants and archaea are for purposes of energy, animal vision converts light into information that the brain processes [14].

The structure of the mammalian visual system can be divided into parts with varying degrees of interconnectivity between them. Visual processing is initiated at the retina, an approximately hemispherical sheet of cells at the back of the eye that are specialized for sensing light. The retina is composed of several cell types, including photo-receptors which have graded responses to visual signals, as well as ganglion cells that receive inputs from the photo-receptors and send all-or-none signals called action potentials down the optic nerve. The rod photoreceptors are especially sensitive to the intensity of light (how bright it is), whereas the cone photoreceptors are superior in registering light's frequency (what color it is). After reception by the retina, light signals pass thru the optic nerve to reach the lateral geniculate nucleus (LGN), which is a pair of layered groups of neurons that each receives information from both eyes. The neurons in the LGN are specialized to respond to visual signals, and establish the so-called "receptive fields." A receptive field is a group of neurons whose activity corresponds to the presence or absence of light with a particular spatial pattern that is special for that receptive field. After passing through the LGN, light signals finally reach the visual cortex for higher-order processing of global visual information and integration with distinct brain areas [13]. A diagram of the visual system's anatomy, described broadly above, is schematized in figure 1.

Through the manifestations of the written word, mathematics, and other symbolic communication, hu-

mans are arguably the most evolved species on Earth in the realm of visual information-processing. These technologies of abstract information representation have enabled us to develop, among other things, corrective eye wear: a device for improving visual acuity even further! However, there are flaws in the human visual system that are less superficial than structural imperfections of the eye itself. One such flaw is inherently tied to the fact that we have two eyes, but only one brain. Thus, if the left eye were shown an entirely different image than the right, the brain would have to somehow compromise on what to actually perceive. How the brain makes this compromise is the primary focus of this work.

Suppose you were presented with figure 2. Specifically, imagine that the left image was shown exclusively to your left eye and that the right image was shown only to your right eye. The phenomenon you experience next is called "binocular rivalry."

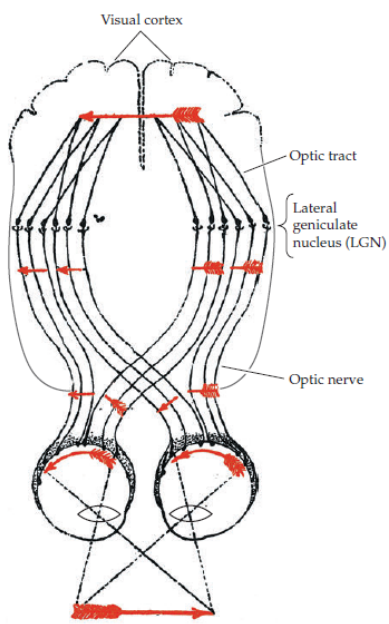


Figure 1: Schematic diagram of the visual system, with the image of a red arrow projected onto various parts of the substrates that would process such an image. This drawing is by Ramón y Cajal from 1892, and was taken from the textbook by Nicholls *et al.* (2001) [13].

Figure 2: Example of gratings typically used in binocular rivalry experiments.

The first scholar [16] to publish research on binocular rivalry was Charles Wheatstone in 1838. He discovered that perception can alternate in time, and the duration that a given perception is dominant depends on the illumination of the object. Brighter images are perceived for longer periods of time than dim ones. Wheatstone influenced physicist Hermann von Helmholtz to study in 1896 binocular rivalry, as well as vision more broadly [6, 7]. Helmholtz claimed that, if he paid attention to an image, it would be dominant in his perception longer. One way he would do so is by actively counting the lines in the grating he was looking at. In 1930, Meenes published research stating that alternations in perception don't always occur in a unitary fashion. In other words, sometimes the images mix together in a patchy way rather than completely switch all at once from one to the other. Interestingly, in 1956 Engel [5] showed that the emotional content of an image can influence its dominance duration. For example, if one was shown a family member in one eye and a uniform white screen in the other, their percept will likely dwell on the image of the family member.

1.1 Two Types of Binocular Rivalry

Much of the previous research on binocular rivalry has tried to explain the origin of perceptual alternations. The difficulty in studying this subject arises from the subjective nature of perception, but also the complex organization of the brain that makes it hard to identify the particular loci at which perceptual rivalry is resolved. There is an ongoing debate on whether rivalry

happens between neural populations representing different images or between populations associated with each eye. The latter hypothesis is referred to as eye-based rivalry and has been the traditional view in the field since the study done by Blake *et al.* in 1979 [1]. In this study, they asked participants to indicate whenever their perception became dominated by one image over the other, at which point the two images would get swapped between the eyes. They observed that, at swap times, the perception also changed, which is consistent with eye-based dominance. The first hypothesis is often referred to as "stimulus-based rivalry" and has gained more support recently. This is because new experimental paradigms have been developed which gave rise to novel evidence in support of this hypothesis. One example is the Flicker & Swap (F&S) paradigm developed by Logothetis *et al.* in 1996 [12]. In this study, subjects were shown two orthogonal gratings to each eye. The gratings flickered on and off at a frequency of 18 Hz and were swapped between the eyes at a frequency of 1.5 Hz. The experiment showed that subjects experienced perceptual alternations where the dominance duration of each grating lasted for about 2 seconds on average. This result is inconsistent with eye-based rivalry, which predicts that perceptual alternations should happen on the same timescale as the swapping (that is less than 1 second).

While the two hypotheses may seem contradictory on the first impression, the evidence for both of them has been given by inherently different experimental frameworks. For example, the results of the Flicker & Swap paradigm hinge critically on the temporal characteristics of the experiment and the frequencies at which the images flicker and swap [9]. On the other hand, traditional rivalry experiments are mainly based on static stimuli that are sometimes swapped between the two eyes at a much lower frequency (1 or 2 swaps per minute). Hence, it is plausible to assume that the two frameworks may be probing two modes of neural competition that happen at different levels in the brain. For example, neural networks in early part of the visual pathway (e.g. the primary visual cortex) may not be dynamically equipped to compete in response to a temporally rich procedure like the F&S, which may cause binocular rivalry to emerge at a higher stage in the visual pathway. This is also reasonable since, moving up the visual hierarchy, representations become more affected by stimulus features rather than monocular input, which might explain the emergence of stimulus-based rivalry in the F&S paradigm. In summary, the two hypotheses about binocular rivalry may not be necessarily contradicting. Indeed, one important objective as we develop a model for rivalry is the reconciliation of those two hypotheses.

1.2 Characteristics of Binocular Rivalry

Another aspect of binocular rivalry that has been extensively studied in the past is the characteristics and features governing its temporal and spatial dynamics. Because of their importance in judging the validity of any computational model of rivalry, we hereby summarize some of the key experimental results about these characteristics.

- Small differences along any dimension in monocular images (e.g. different orientations, colors, etc.) are enough to trigger perceptual rivalry.
- Perceptual alternations is the hallmark of binocular rivalry and the behavior of these alternations depends on many parameters such as light settings, stimulus characteristic, emotional state, etc.
- For perceptual oscillations to occur, dissimilar images must be presented to the eyes for long enough periods of time (on the order of a hundred milliseconds). Those oscillations can be disrupted by transient changes to the environment. As a result, dynamical trajectories of rivalry include both a transient and a steady-state components.
- Changing stimulus strength, e.g. contrast or brightness, affects the perceptual dominance of the different images.

The last point was studied thoroughly by W. J. M. Levelt, who published his results in a landmark 1965 dissertation [10]. Levelt's experiments yielded four main propositions that elucidate the link between stimulus strength and the duration of perceived stimuli. A computational model of rivalry can therefore be tested against these "laws" of binocular rivalry to determine its validity.

1.3 The Wilson model

One of the most important models of binocular rivalry was developed by Hugh Wilson in 2003 [18]. With this model, Wilson tried to reconcile the two *prima facie* contradictory hypotheses mentioned above: eye-based and stimulus-based rivalry. While previous models were able to generate trajectories similar to those observed in traditional rivalry (i.e. rivalry based on monocular competition), they were not able to explain the results shown by procedures such as the F&S. Wilson's idea was to construct a two-layer hierarchical network to mimic the organization of the brain (see Figure 3). Rivalry is then assumed to be resolved at the higher level in the network, where input is not exclusively monocular. The lower level of Wilson's model consisted of neuronal populations that are driven purely by monocular input, while the higher level consisted of populations that pooled their input from the lower level populations of both eyes. Each population in the higher level was selective for a specific stimulus so its input was pooled from lower level neurons that responded only to that stimulus. Amazingly, when the model was simulated under the experimental conditions of traditional and F&S rivalry, it was able to generate trajectories consistent with both results. This provided evidence that both eye-based and stimulus-based rivalry can be performed by the same network. However, for this to happen, such a network must be hierarchically organized and have two layers at minimum.

Our approach in this paper is to start from minimal assumptions about the structure of a candidate binocular rivalry model, and build it up sequentially such that its behavior is consistent with experimental results. That way, we can understand the role each part plays in the final behavior of the model, and hopefully establish a link between those parts and potential biological mechanisms of rivalry. We draw inspiration from the Wilson model and we build on very simple models presented by S. Strogatz in ref. [15] in the following sections.

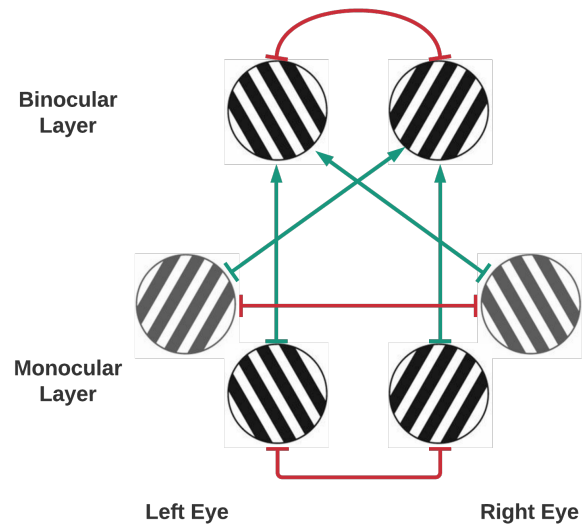


Figure 3: Hierarchical architecture of the Wilson model [18]. Red lines correspond to mutually inhibitory connections, while green lines correspond to excitatory synaptic inputs. In each eye, there are two populations, each selective to a particular input to that specific eye. Populations in the monocular layer then connect to populations in the higher layer, which get inputs from both eyes but respond only to one type of stimulus.

2 Models without Adaptation

2.1 Symmetric Populations

Consider the system defined in problem 8.1.14 of [15]:

$$\begin{aligned} \dot{x}_1 &= -x_1 + F(I - bx_2) \\ \dot{x}_2 &= -x_2 + F(I - bx_1) \end{aligned} \tag{1}$$

where $b, I \geq 0$ are parameters, x_1 and x_2 are the average firing rates of two populations of neurons, also inherently non-negative quantities. The function $F(x) = 1/(1 + e^{-x})$ couples the neuronal activities through mutual inhibition. The parameter b encapsulates the strength of the mutual antagonism between the two neuron populations. The parameter I represents the input stimulus strength, which involves properties of the relevant image in the context of binocular rivalry.

The appearance of the phase plane for this system depends on the parameters I and b. However, no matter the particular values of b and I, there are always exactly 1 or 3 intersections of the nullclines. The intersections of the nullclines (x_1^*, x_2^*) represent fixed

points, given by the solutions to the algebraic system:

$$x_1^* = F(I - bx_2^*)$$
$$x_2^* = F(I - bx_1^*)$$

These are transcendental equations that are most readily illuminated via graphical means. For an interactive graph that shows the intersections of the nullclines for various parameter values, follow this link. The nullclines are shown here for fixed $I = 0$ and varying b, in figures 4, 5 and 6. This sequence of plots shows that a pitchfork bifurcation occurs at a critical value of $b = b_c \approx 5.87$ when $I = 0$. This pitchfork bifurcation is shown to be supercritical via numerical means in figure 7.

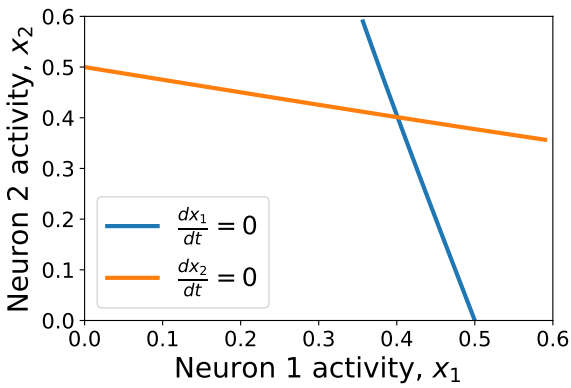


Figure 4: Nullclines of system (1) when $b = 1$, $I = 0$. The symmetric equilibrium $x_1^* = x_2^*$ is stable here.

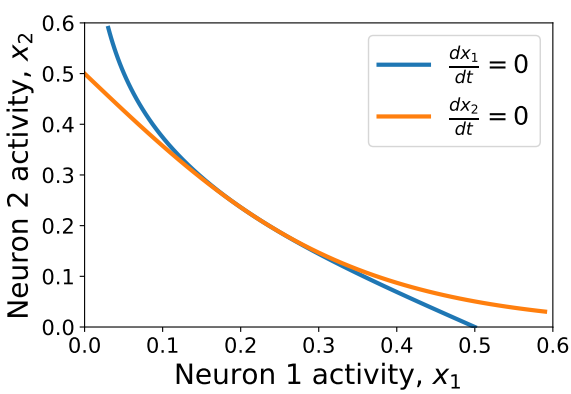


Figure 5: Nullclines of system (1) at a pitchfork bifurcation when $b = b_c \approx 5.87$, $I = 0$. The bifurcation occurs when the neuronal steady-states are equal $x_1^* = x_2^* \approx 0.218$.

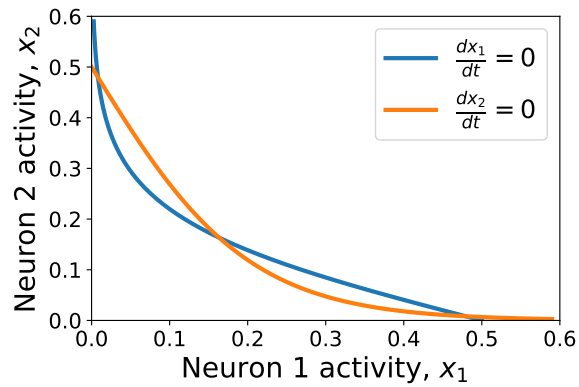


Figure 6: Nullclines of system (1) when $b = 10$, $I = 0$. The system is bi-stable in this regime, as confirmed by figure 7.

The symmetric fixed point $x_1^* = x_2^* = x^*$ is always a solution to (1), since the equation

$$0 = -x^* + F(I - bx^*)$$
$$x^* = \frac{1}{1 + e^{bx^* - I}}$$

always has a solution. This is because the number $b \geq 0$ only affects the steepness of the sigmoid function F while $I \geq 0$ only shifts it horizontally. No matter their values, there always exists an intersection between the 45° line $y = x$ and $y = F(I - bx)$ in the positive xy-plane. Evidence for this is provided at this link. Linearizing the nonlinear system (1) about the symmetric equilibrium $x_1^* = x_2^* = x^*$ yields a stability matrix with trace $\tau = -2$ and determinant $\Delta = 1 - b^2(x^*)^4(e^{bx^* - I})^2$. The sign of Δ determines whether or not the symmetric fixed point is stable or not. Since the entire second term is always positive, it is easy to see that we would have $\Delta < 0$ for sufficiently large b. It is also clear that Δ gets more negative as I goes down. Indeed, for an appropriately chosen b, a pitchfork bifurcation occurs for I in this model in the opposite sense of how it is occurs for b. In other words, the pitchforks in the bifurcation diagrams of these parameters point in opposite directions.

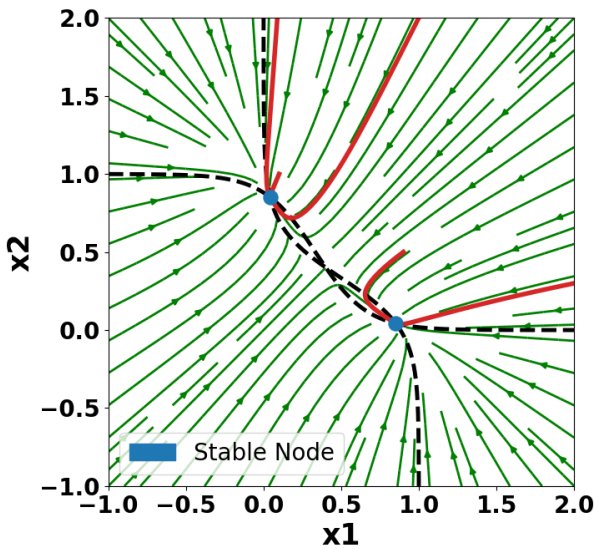


Figure 7: Phase portrait of system (1) when $b = 6$, $I = 2$. The horizontal axis represents x_1 and the vertical represents x_2. The nullclines are shown with the dotted curves. The vector field is shown with the green arrows and sample trajectories are shown in red. The system is bi-stable in this regime. The 45°-line $x_2 = x_1$ separates steady states with $x_2 > x_1$ from those with $x_1 > x_2$ (it is the "separatrix").

A physiological interpretation of figures 4, 5, and 6 is as follows: when the stimulus or mutual antagonism are not strong enough (as in figure 4), neither neuronal population is more active than the other and they settle to the symmetric equilibrium where both of their activities are the same. As the mutual antagonism is increased while the stimulus is held constant, this symmetric equilibrium loses stability in a pitchfork bifurcation and creates the separatrix $x_1 = x_2$. This model therefore predicts that in an idealized scenario without noise, initial conditions with $x_1 = x_2$ will tend to asymptotic states with $x_1 = x_2$. If there is the slightest deviation, however, the population that is more active initially will be much more active than the other in the long run. Importantly, this dynamical system does not have limit cycles. This is proven by noting that the divergence of its vector field:

$$\frac{\partial \dot{x_1}}{\partial x_1} + \frac{\partial \dot{x_2}}{\partial x_2} = -2$$

is not identically zero and does not change sign on the plane. Because of this, Bendixson's criterion implies that this system cannot possess limit cycles [8]. This is a flaw of this simple model of binocular rivalry, since stable oscillations are known to occur in the visual system. This phenomenon is referred to as "perceptual periodicity" in ref. [3]. In order to account for these oscillations in perception, the neuronal populations controlling which eye's image is perceived need to get tired after a while. In other words, there must be an "adaptation time," as discussed in subsection 2.3. In the next subsection, we consider a variant of the simple model where the symmetry is broken.

2.2 Asymmetric Inputs

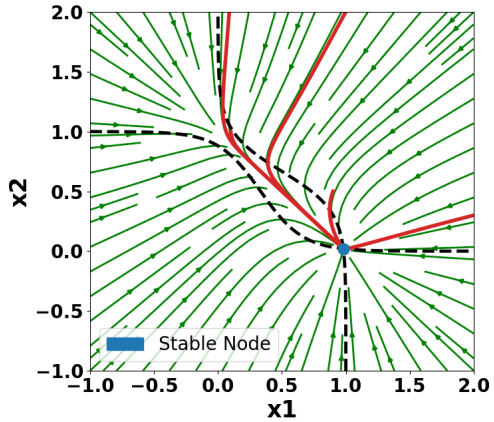
We now consider the asymmetric version of the model described above. The motivation here is to see if, with a simple model such as this, we can replicate qualitative features of the relationship between perceptual dominance and stimulus strength. Varying stimulus strength between the two stimuli corresponds to making I and/or b different for the two populations. For example, in the case where we vary I, the model becomes:

$$\dot{x_1} = -x_1 + F(I_1 - bx_2)$$
$$\dot{x_2} = -x_2 + F(I_2 - bx_1)$$

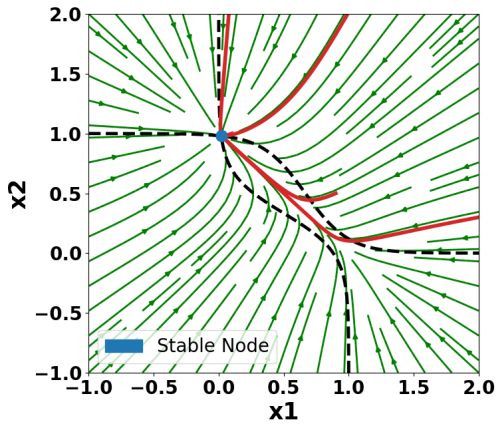
where $b, I_1, I_2 \geq 0$. In this case, we look at how different values of I_1 and I_2 affect the phase portrait in terms of the number and symmetry of fixed points as well as the type of bifurcations that occur as the parameter b is varied.

Effect of Breaking the Symmetry Near the Bifurcation Point

Consider the symmetric bi-stable system (shown in section 1) immediately after the bifurcation occurs. There, the system has two symmetrically-located stable equilibria with one unstable point in the middle, as shown in figure 7. We are interested in how varying I_1 and I_2 for the two different populations affects this symmetry. Figure 8 shows how the system transitions from the bi-stability to uni-stability with one fixed point whose location depends on the difference between I_1 and I_2. We also examined how fixing I for the two populations and varying b_1 and b_2 affects bistability at the bifurcation borderline. In both cases, breaking the symmetry results in the disappearance of bistability. However, in the case where $I_1 \neq I_2$, the

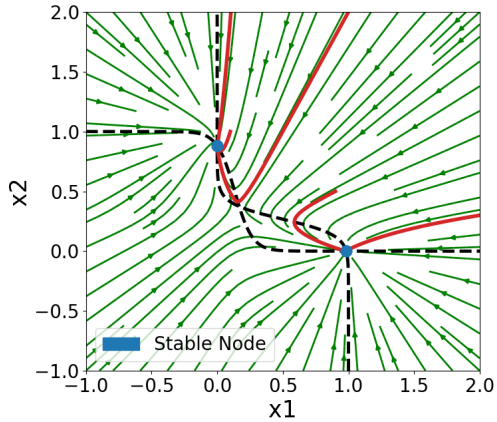


(a) $b = 6, I_1 > I_2$

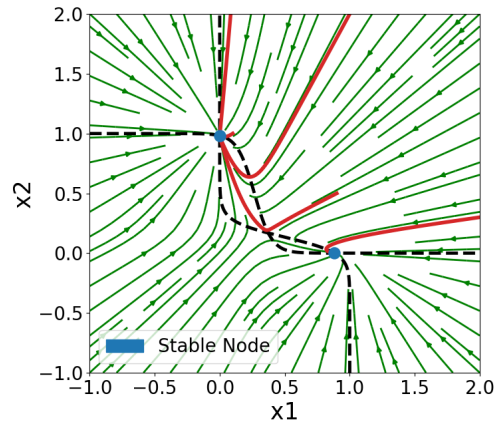


(b) $b = 6, I_1 < I_2$

Figure 8: Effect of varying I_1 and I_2 on the bi-stable system. Red traces represent example trajectories.



(a) $b = 15, I_1 > I_2$



(b) $b = 15, I_1 < I_2$

Figure 9: Effect of varying I_1 and I_2 on the bi-stable system beyond the bifurcation point. Red traces represent example trajectories.

new monostable fixed point is shifted towards the population with the higher I, opposite to the case where I is fixed and $b_1 \neq b_2$; in that case, the monostable point shifts to the population with the lower b value.

Effect of Breaking the Symmetry Beyond the Bifurcation Point

We also considered how breaking the symmetry well beyond the bifurcation point affects the system's bi-stability. We varied the parameters in a similar fashion as in Figure 8, but in a range of values that places the system well beyond the bifurcation point. In this case, breaking the symmetry does not result in the disappearance of bi-stability. However, the locations of the two stable fixed points become asymmetrical. For example, in the case where the stimulus strength is allowed to vary between the two populations, the steady-state activity of the population with the higher I is stronger than that of the population with the lower I. These effects are show in figure 9.

Models with Adaptation

2.3 Symmetric Populations

Consider the model defined in problem 8.2.17 of [15]:

$$\begin{aligned}\dot{x}_1 &= -x_1 + F(I - bx_2 - gy_1)\\ \dot{y}_1 &= (-y_1 + x_1)/T\\ \dot{x}_2 &= -x_2 + F(I - bx_1 - gy_2)\\ \dot{y}_2 &= (-y_2 + x_2)/T\end{aligned} \tag{2}$$

where, as before, $F(x) = 1/(1+e^{-x})$ and parameters b and I represent the mutual antagonism and stimulus strength, respectively. The new variables y_1 and y_2 represent the adaptation of their corresponding neuronal populations on a time scale $T > 0$. These numbers feed into the gain function F, shifting it horizontally with strength $g > 0$.

If we set $x_1 = y_1 \equiv u$ and $x_2 = y_2 \equiv v$, this yields $\dot{y}_1 \equiv 0$ and $\dot{y}_2 \equiv 0$, as well as the nullclines

$$\begin{cases}0 &= -u + F(I - bv - gu)\\ 0 &= -v + F(I - bu - gv)\end{cases}$$

$$\begin{cases}u &= F(I - bv - gu)\\ v &= F(I - bu - gv)\end{cases}$$

The above system of algebraic solutions always has the fixed point $x_1^* = x_2^* = y_1^* = y_2^* \equiv u$. This is true by identical reasoning as employed above for the 2-dimensional model. The line going through the origin with a slope of 1 ($y = x$) always intersects the sigmoidal curve $y = F(I - (b - g)x)$ at least once, regardless of I,b, or g. This is because I shifts the sigmoid horizontally, while b and g both modulate its slope. Neither of these tuning parameters transform the sigmoid in a way that makes it escape intersecting the 45-degree line at least once. An alternative argument is that the two sigmoids $u = F(I - bv - gu)$ and $v = F(I - bu - gv)$ always intersect at least once. Graphical evidence for this claim is provided here.

The stability matrix of this model evaluated at the fixed point u is given by

$$\begin{bmatrix}-1 & -gF' & -bF' & 0\\ 1/T & -1/T & 0 & 0\\ -bF' & 0 & -1 & -gF'\\ 0 & 0 & 1/T & -1/T\end{bmatrix}$$

where $F' = F'(I - bu - gu)$. If we let $c_1 = 1$, $c_2 = gF'$, $c_3 = bF'$, and $d_1 = 1/T$, then this matrix can be written in the form

$$\begin{bmatrix}-c_1 & -c_2 & -c_3 & 0\\ d_1 & -d_1 & 0 & 0\\ -c_3 & 0 & -c_1 & -c_2\\ 0 & 0 & d_1 & -d_1\end{bmatrix}$$

which may be put in block matrix form

$$\begin{bmatrix}A & B\\ B & A\end{bmatrix}$$

where

$$A = \begin{bmatrix}-c_1 & -c_2\\ d_1 & -d_1\end{bmatrix}$$

$$B = \begin{bmatrix}-c_3 & 0\\ 0 & 0\end{bmatrix}$$

We may now state the characteristic equation

$$\begin{vmatrix}-c_1 - \lambda & -c_2 & -c_3 & 0\\ d_1 & -d_1 - \lambda & 0 & 0\\ -c_3 & 0 & -c_1 - \lambda & -c_2\\ 0 & 0 & d_1 & -d_1 - \lambda\end{vmatrix} = 0$$

in block matrix form as

$$\begin{vmatrix}A - I\lambda & B\\ B & A - I\lambda\end{vmatrix} = 0$$

which is identical to the equality

$$|A - B - I\lambda||A + B - I\lambda| = 0$$

This indicates that the eigenvalues of the 4×4 matrix above are the same as the eigenvalues of the 2 matrices $A + B$ and $A - B$. The matrix $A + B$ has trace $\tau = -c_1 - c_3 - d_1 < 0$ and determinant $\Delta = d_1(c_1 + c_2 + c_3) > 0$. Since we have the relation $\tau = \lambda_1 + \lambda_2$ and $\Delta = \lambda_1\lambda_2$, both eigenvalues λ_1 and λ_2 of $A+B$ must be negative. Additionally, the matrix $A - B$ is given by

$$\begin{bmatrix}-c_1 + c_3 & -c_2\\ d_1 & -d_1\end{bmatrix}$$

and has trace $\tau = c_3 - c_1 - d_1$ and determinant $\Delta = d_1(c_1 + c_2 - c_3)$. Putting this back in terms of the original parameters, the trace is $\tau = bF' - 1 - 1/T$ and

the determinant is $\Delta = (1 + gF' - bF')/T$. Either of these quantities could be negative, depending on the parameter values. This suggests that the symmetric fixed point u could lose stability in more than one way. The bifurcation diagram in figure 10 shows that a supercritical pitchfork bifurcation can be induced by modulating the stimulus strength I.

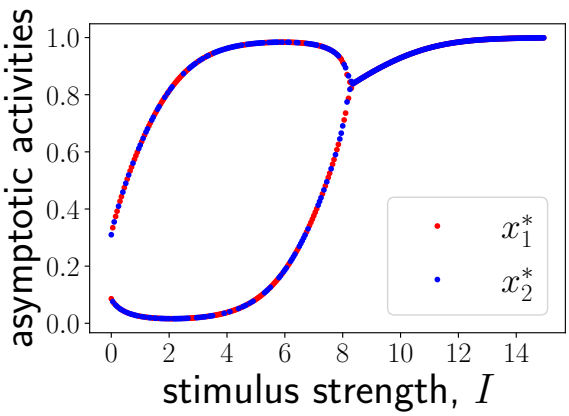


Figure 10: Supercritical pitchfork in the system given by equation 2 for parameter values $g = 0.5$, $b = 7.5$, and $T = 1$. The bifurcation occurs around $I = 8$.

The fixed point u is also shown to lose stability in a supercritical Hopf bifurcation as the adaptation strength g is increased in figure 11. This proves that the 4-D model can account for perceptual alternations.

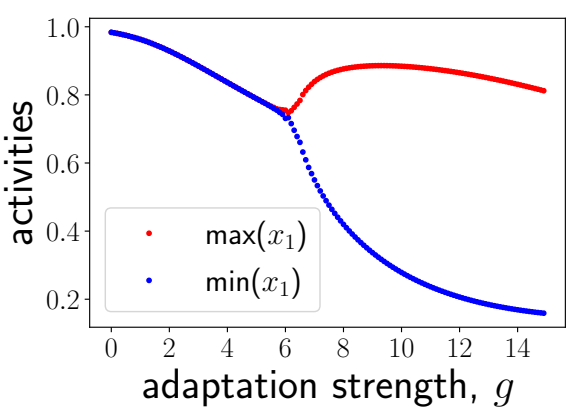


Figure 11: Supercritical Hopf bifurcation in the system given by equation 2 for parameter values $I = 10$, $b = 6$, and $T = 5$. The bifurcation occurs around $g = 6$.

3 Results

Consider the extension of the previous model where distinct inputs I_1 and I_2 are provided to each population:

$$\begin{aligned}
\dot{x}_1 &= -x_1 + F(I_1 - bx_2 - gy_1) \\
\dot{y}_1 &= (-y_1 + x_1)/T \\
\dot{x}_2 &= -x_2 + F(I_2 - bx_1 - gy_2) \\
\dot{y}_2 &= (-y_2 + x_2)/T
\end{aligned}$$

This system is solved numerically in figures 12, 13 and 14. These results show that 3 distinct dynamical modes identified in the recent literature [4] can be reproduced with this simple model.

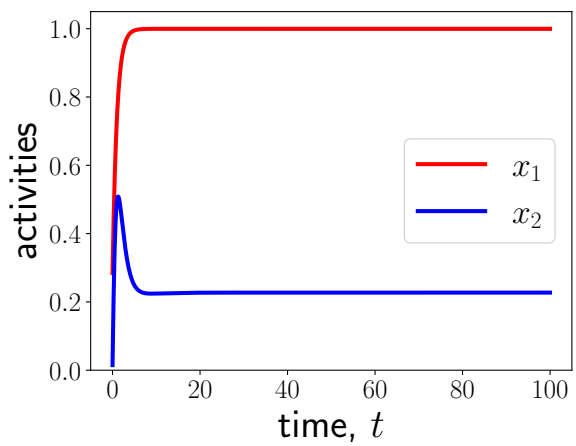


Figure 12: Winner take all. Parameter values are $I_1 = 10$, $I_2 = 9$, $g = 1$, $b = 6$, and $T = 5$.

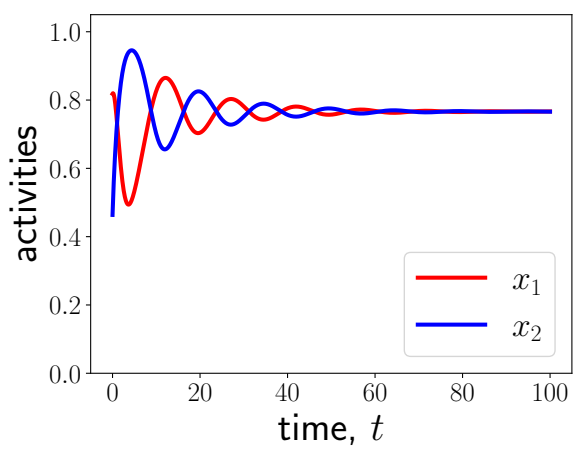


Figure 13: Simultaneous activity. Parameter values are $I_1 = I_2 = 10$, $g = 5.5$, $b = 6$, and $T = 5$.

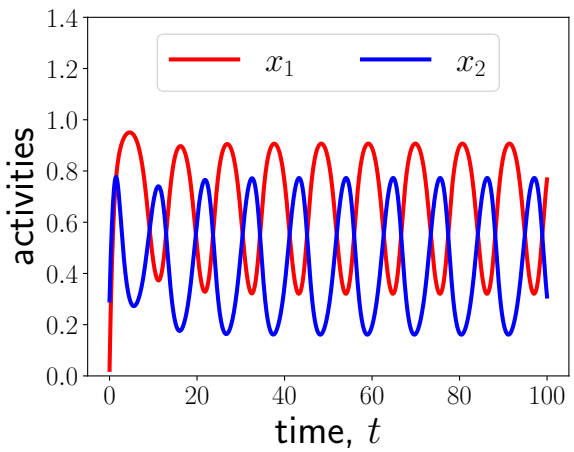


Figure 14: Rivalry oscillations. Parameter values are $I_1 = 10$ $I_2 = 8$, $g = 10$, $b = 6$, and $T = 5$.

The "winner take all" motif shown in figure 12 corresponds to one perception quickly dominating over the other. The outcome shown $x_1^* > x_2^*$ is due to the inequality of inputs $I_1 > I_2$ and can be interpreted as the static image presented to population 1 being brighter than that presented to population 2. This interpretation would be consistent with one of Wheatstone's discoveries mentioned in the introduction.

"Simultaneous activity" is portrayed in figure 13. The beginning phase of the trajectories $t < 60$ may be interpreted as a duration of time where perceptions are mixing dynamically in a patch-like fashion. The subsequent duration $t > 60$ represents a stable state where features of both images are perceived statically, a phenomenon known as "binocular fusion" [2]. This convergence to equal steady-states only occurs because the stimuli have equal intensities $I_1 = I_2$. If instead we were to increase one of the intensities over the other, the population with the higher stimulus intensity would have a larger steady state.

The third scenario of "rivalry oscillations" is depicted in figure 14, where stable oscillations occur between the populations. The population with the higher stimulus intensity is dominant for longer percentage of the oscillatory period (this is exemplified by the red trace in figure 14).

Two important distinctions between our rivalry oscillations (figure 14) and those of ref. [4] exist. First, the shapes of the activity oscillations in our model look sinusoidal, whereas those in the Darki-Rankin paper reflect the adaptive response of the neuronal populations: there is a fast spike followed by a slow decay to some intermediate activity level before dropping down to zero (see the green traces in figure 3c of their paper). This issue can be resolved by re-interpreting our variables x_1 and x_2 as the phases of the activities of the respective populations, instead of the absolute activities themselves. Otherwise, we would need to include a 3-dimensional separation of timescales for each population to get the shapes correct, as Darki and Rankin have done. Secondly, the period of the rivalry oscillations is larger than the period of simultaneous activity oscillations in Darki and Rankin's work. Whereas for our model, the period of damped oscillations in the simulataneous activity case appears equal to that of the limit cycle in the rivalry oscillations case. This is due to the fact that our model does not possess a global bifurcation like that of the Darki-Rankin model.

4 Discussion

In this paper, we have attempted to construct a model of binocular rivalry with minimal complexity. Our objective in this process was to capture the most important features of binocular rivalry such as perceptual alternations and stimulus dependence. Here we outline some of the limitations and successes of our final model.

Features of Binocular Rivalry Captured by the Adaptation based 4 Dimensional Model

As shown in figure 14, with the correct parameter settings, our model is able to produce perceptual alternations similar to what is observed experimentally. These alternations emerge as a direct consequence of including adaptation in the model. We have proved that without a slow process, namely adaptation, that causes neuronal activities to self-decay, oscillations cannot occur (see section 2.1). This is in congruence with what has previously been shown by Wilson [18] and confirms the importance of neural adaptation in binocular rivalry. Through a bifurcation analysis, we have also shown that a Hopf bifurcation occurs around a value of $g = 6$, which corresponds to the point at which adaptation is strong enough to allow for perceptual oscillations.

The results of our model for asymmetric values of stimulus strength (figure 14) also agrees with Levelt's propositions relating stimulus strength to the average dominance duration of each stimulus [10]. According to Levelt, as the strength of one stimulus increases, the average duration for which it is perceived also increases. The average duration is calculated by averaging the lengths of the intervals during which that stimulus is perceived. In figure 14, those intervals correspond to the intervals where x_1 (red) $> x_2$ (blue). As we can see, the average duration of x_1 is higher than that of x_2, consistent with I_1 being higher than I_2.

Limitations of the Model

One of the limitations of this model is that it fails to capture the biologically-accurate shape of each neural response. The activity of real neuronal populations include a high transient at the beginning, corresponding to aggregated initial spikes, which decay rapidly to moderate levels and then slowly die off. This behavior implies a separation of timescales which is not implemented in our model. It is known that the activity timescales of inhibitory and excitatory populations in the brain are different [17]. Therefore, adding separate equations for inhibitory populations with a different time scale could result in trajectories that look more like biological neuronal activities.

Although not investigated here, it is suspected that our model cannot reproduce the dynamics of flicker & swap experiments where the period of oscillations in response exceeds that of the period with which images are swapped between eyes. This experimental result embodies a nuanced aspect of binocular rivalry that appears to depend on multiple layers of neurons communicating, which our model does not include.

Future Work

So far, we have only considered models that are autonomously driven, i.e. there is no forcing input. One scenario we could consider in the future is what happens to the behavior of the neurons when use their activities are driven by time dependent inputs. This time dependence can correspond to time-varying intensities or periodic forcing. In the periodic case, for example, stimuli can be swapped between the two eyes at a some frequency, or they could flicker on and off, or both. These types of input have been explored in previous models [4, 18], which were able to show results similar to those obtained experimentally with the F&S procedure. Testing our model under similar conditions may be helpful in understanding more about its limitations and possible improvements.

One of the limitations of Wilson's model is that the activities of monocular populations are unaffected throughout the course of perceptual rivalry. This is a consequence of the absence of feedback, which violates what we know about the abundance and importance of feedback connections in the brain. Therefore, another direction to explore would be to build a hierarchical network based on this model but with feedback. This feedback, for example, could represent attentional modulation, which has been implemented successfully as a model for binocular rivalry [11]. Feedback allows higher level populations to affect monocular population activities, which can in turn contribute to different, perhaps richer, dynamic behaviors. With the inclusion of feedback, one can also incorporate the framework of predictive coding, a highly popular theory for how the brain learns its sensory representations.

5 Methods

All computational simulations were carried out in Python, using either the forward Euler method or SciPy's built-in solver "odeint."

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The website of the renowned psychology professor and binocular rivalry expert Randolph Blake was an invaluable resource for information on the historical context of the subject summarized here.

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